

Boundary-Relative State-Time

Measurement Lineage Across Nested System Boundaries

Article type: Research Article

Abstract

Elapsed laboratory time does not uniquely determine how much a system has evolved. This article proposes boundary-relative state-time as a systems-measurement framework for accumulated histories indexed by explicit analytical boundaries. Its primary contribution is a measurement-lineage rule: when a boundary is enlarged at inherited resolution, contained histories and their measurement semantics must remain recoverable and coherent; otherwise the larger model is a replacement rather than an extension. The framework separates direct full-boundary accumulation, component-preserving ledgers, and optional scalarization, and distinguishes analytical boundary expansion from coarse-graining or physical modification. Geometric and event-count examples show nonempty admissible constructions and pre-outcome rejection criteria, while an LED thermal-coupling case illustrates boundary reclassification without claiming predictive validation. Empirical usefulness is reserved for prospective tests of pre-specified implementations against established domain metrics.

Keywords: system boundary; state-time; accumulated history; measurement lineage; nested-boundary composition; coarse-graining; systems theory

Practitioner Points

- Expanding a system boundary should not silently change the calibrated meaning of the history already assigned to a contained subsystem.
- Preserve heterogeneous component histories before aggregation; a direct full-boundary scalar and a ledger scalarization answer different questions.
- Distinguish boundary expansion from coarse-graining and model replacement so changes in reported history remain traceable to physics, resolution, or measurement lineage.

1. Introduction

Clock time answers when and for how long a system exists or is observed. It does not, by itself, determine how much the system has evolved. Two systems can experience the same laboratory duration while undergoing very different thermal, structural, chemical, electrical, or computational trajectories. Conversely, two systems can arrive at similar measured endpoints through histories that differ in the amount, direction, and rate of internal change. A sufficiently specified bounded system may therefore admit one or more accumulated internal-evolution coordinates that need not advance uniformly with external laboratory time; the question developed here is how such coordinates remain interpretable when the analytical boundary changes.

This distinction is not unprecedented. Intrinsic or material time has been proposed in aging and dissipative systems [1-4]. Thermodynamic length treats transformations as metric distances between thermodynamic states and links path geometry to dissipation [5]. Information length describes accumulated motion through statistical-state space [6,7]. Stochastic thermodynamics uses dynamical activity and related kinetic quantities to count or weight trajectory-level changes and derive speed limits and uncertainty relations [8-10]. These lines of work establish that elapsed laboratory time is not the only meaningful temporal coordinate for evolving systems.

The proposal does not claim novelty for state-space trajectories, internal or additive clocks, activity measures, system boundaries, open-system composition, functorial restriction, or boundary energy accounting individually. Its narrower contribution is an accumulated-history measurement-lineage architecture for nested analytical boundaries:

once a subsystem history and its measurement semantics are declared, enlargement at inherited resolution must preserve their recoverability and meaning if the larger description is to count as an extension of the same analysis. Added histories may be retained without forcing them into one scalar, and a changed inherited rule is treated as a new model lineage rather than as an effect of boundary movement. The boundary is therefore prior to any chosen metric or accumulation rule.

The framework therefore separates six questions that are often collapsed: (i) What is the system boundary? (ii) What variables define the system's state for the problem of interest? (iii) What energy, matter, information, and control flows cross the boundary in either direction? (iv) What transitions are accessible from the current state under the present environment and coupling? (v) By what pre-specified rule should the resulting internal evolution be accumulated relative to external time? and (vi) if the boundary is expanded or contracted, what parts of the previous accounting must remain invariant?

State-time does not replace laboratory time, energy, entropy, accumulated dose, damage measures, or established domain metrics, and no single equation defines it. Implementations may use weighted path length, event counts, thermodynamic or information-geometric distance, endochronic measures, or another pre-specified domain rule. The contribution is narrower: preserve measurement lineage as nested analytical boundaries change. Restriction maps, ledgers, and aggregation rules make that requirement operational; the examples show that admissible constructions exist and can reject some alternatives before outcome fitting; prospective studies test whether an implementation adds value beyond established metrics.

2. Boundary-First Definitions

Choose the boundary B before interpreting quantities inside it. The boundary may be physical, functional, or analytical. 'Bounded' does not mean closed: the framework also applies to open systems, so relevant exchanges across B must be recorded. A thermodynamically closed system excludes matter transfer; an isolated system excludes both matter and energy transfer.

Accounting across B is bidirectional. Energy, matter, and information may enter, leave, or be redirected, while the environment can change which internal pathways are available. Expanding B can turn a former boundary flux into an internal transfer without changing the physical event.

For a chosen boundary B, let $x_B(t)$ be the state vector containing the variables judged relevant to the question. The state vector is not required to include every microscopic degree of freedom. It is an operational model. A thermal aging problem may use temperatures, gradients, and structural markers; an electrical reliability problem may use current, voltage, junction temperature, and degradation observables; a computational problem may use activation, memory, or routing states.

Because the components of x_B can have different physical dimensions, define normalized coordinates z_B using reference scales s_i :

$$z_i(t) = [x_i(t) - x_{i,ref}] / s_i \quad (1)$$

The reference scales may represent measurement resolution, engineering tolerance, a failure threshold, a natural fluctuation scale, or another domain-specific rule scale, and must be declared within M_B . Here “rule calibration” means the declared parameterization and reference semantics of M_B - scales, weights, event definitions, normalization, and observational conventions - not VIM metrological calibration unless traceability is explicitly documented. A valid implementation must describe the same physical history consistently under coordinate change. For a geometric implementation, W_B therefore transforms as a covariant weighting tensor and, when positive definite, as a Riemannian metric tensor. Nested-boundary inherited-block tests are evaluated relative to the declared subsystem projection or splitting: reparameterizations preserving that projection may change coordinates but not the inherited bilinear form; arbitrary mixing of inherited and newly added coordinates changes the decomposition and is a model change.

2.1. Core quantities

Symbol	Meaning	Role	Notes
B	Chosen system boundary	Defines what is counted	Changing B can reclassify a flow as external or internal.
t	External/laboratory time	Reference coordinate	Measured by ordinary clocks.
$z_B(t)$	Normalized state vector	Locates the system in state space	Domain-specific variables and scales.
$\gamma_B^{\text{full}}(t)$	Direct full-boundary rate density (when defined)	Rate density for the absolutely continuous subclass of direct scalar histories	Not required for jump/singular additive histories; not identical to power or energy flux.
ξ_B^{full}	Direct full-boundary state-time	Scalar accumulated history from M_B applied to the complete state at boundary B	May include coupled motion/cross terms; need not equal a scalarization of Ξ_B .
Φ_E	Energy flux across boundary	Driver/constraint	May include heat, work, electrical, chemical, radiative terms.
$\mathcal{T}_B$	Accessible transition set	Constrains possible evolution	Depends on state, environment, barriers, and coupling.
$u(t)$	Control action	Deliberate intervention	May be internal or external depending on B .
M_B	Measurement / accumulation rule	Defines what relevant change counts	Must be explicit, domain-calibrated, and preferably pre-specified before outcome testing.
Ξ_B	Boundary-composition history ledger	Preserves declared component histories under boundary expansion	Primary composite accounting object; heterogeneous entries remain separate before justified aggregation.
π_B	Projection onto a contained boundary	Recovers embedded subsystem history	Enforces consistency for $B_1 \subset B_2$.
π_{ji}	History projection from B_j to contained B_i	Recovers inherited subsystem trajectory	Must compose coherently across nested boundary chains.
R_{ji}	Measurement-rule restriction	Tests whether M_{B_j} genuinely extends M_{B_i}	Must satisfy $R_{ki} = R_{ji} \circ R_{kj}$ for nested boundaries.
p_{ji}	State-space projection from expanded to inherited boundary	Identifies inherited state coordinates	Used to define fiberwise metric compatibility in product/local descriptions.
c_B	Coarse-graining map	Reduces model resolution	Distinct from boundary projection; may intentionally discard information.
$\mathcal{H}(B)$	History-data assignment at boundary B	Stores admissible histories/records	Restriction maps π_{ji} form a contravariant functor / Set-valued presheaf over nested boundaries.
$\mathcal{M}(B)$	Measurement-rule assignment at boundary B	Stores admissible accumulation rules	Restriction maps R_{ji} preserve inherited measurement semantics as a contravariant rule family.

Symbol	Meaning	Role	Notes
σ_B	Optional ledger scalarization	Scalar summary $S_B(\Xi_B)$ of declared ledger components	Requires an explicit compatible aggregation rule; generally distinct from ξ_B^{full} .
$\mathcal{L}(B)$	Ledger domain at boundary B	Stores labeled component-preserving ledger objects	Heterogeneous entries need not share a common algebraic structure.
$\mathcal{S}(B)$	Scalarization-rule assignment at boundary B	Stores admissible optional aggregation maps $S_B: \mathcal{L}(B) \rightarrow \mathbb{R}_{\geq 0}$	Used only where ledger aggregation is meaningful; boundary-local by default.
T_{ji}	Aggregation-rule restriction from B_j to B_i	Preserves inherited scalarization semantics when cross-boundary comparison is claimed	Must satisfy identity/composition laws analogously to R_{ji} .

3. From Boundary to State-Time: Theory and Measurement

Let $\Gamma_B[t_0, t_1]$ denote an admissible bounded history record over an interval, and let $\mathcal{H}(B)$ denote the domain of such records. Let $\mathcal{M}_{\text{scalar}}(B)$ denote measurement rules that return a direct scalar accumulated history. For the direct scalar case, define the accumulation map $A_B: \mathcal{H}(B) \times \mathcal{M}_{\text{scalar}}(B) \rightarrow \mathbb{R}_{\geq 0}$ by:

$$\xi_B^{\text{full}}[t_0, t_1] = A_B(\Gamma_B[t_0, t_1]; M_B) \quad (2)$$

A valid direct scalar state-time implementation is nonnegative, history-sensitive, and additive across adjacent intervals under a fixed boundary and rule. Here history-sensitive means that the functional need not be determined solely by terminal state and may distinguish histories that share the same endpoints; it does not require a distinct scalar value for every distinct trajectory. It equals zero whenever the record contains no evolution that M_B declares relevant. In unreduced coordinates, the converse need not hold if M_B explicitly declares some directions irrelevant or null. The rule M_B determines what counts as relevant change and how different changes are weighted. Non-additive but consistently composable native history objects - for example a maximum excursion, multiplicative survival quantity, or stateful damage record - may still be preserved as components of Ξ_B , but they are not denoted ξ_B^{full} unless mapped to an explicitly additive accumulated coordinate. For composite boundaries, ξ_B^{full} is not the same object as the component ledger Ξ_B or an optional ledger scalarization σ_B introduced in Section 3.5.

$$\xi_B^{\text{full}}(t_0, t_1) = \int_{[t_0 \rightarrow t_1]} \gamma_B^{\text{full}}(t) \, dt \quad (3)$$

Equation (2) is the general direct-scalar definition. When the additive functional is absolutely continuous with respect to laboratory time, it admits a rate density $\gamma_B^{\text{full}}(t)$ and may be represented by Equation (3). Equation (3) is therefore a special case, not a requirement on every admissible additive history. Additive functionals may also contain jumps or singular contributions; the threshold-event construction in Section 3.10 is temporally additive when detector state is inherited across interval boundaries but need not possess an ordinary instantaneous rate density. The framework can therefore retain event-count and other additive histories without forcing them into a continuous-rate representation.

3.1. Candidate geometric realization

$$\gamma_B^{\text{full}}(t) = \sqrt{(\dot{z}_B(t))^T W_B[z_B(t), t] \dot{z}_B(t)} \quad (4)$$

Here W_B is a positive-semidefinite weighting tensor; when positive definite, it defines a Riemannian metric tensor. Positive semidefiniteness is permitted only when any null directions are explicitly declared irrelevant under M_B ; equivalently, the induced form must be positive definite on the quotient of state directions regarded as relevant. If every nonzero declared state direction is relevant, W_B must be positive definite. This candidate makes ξ_B^{full} a weighted path length through the complete chosen state description: back-and-forth motion contributes in both

directions even when net endpoint displacement is small, and cross terms may represent coupled motion between state coordinates. It is useful because it is simple and testable, not because weighted arc length is claimed as a new mathematical object.

For discrete observations, the same geometric candidate can be approximated as:

$$\xi_B^{full} \approx \sum_k d_B(z_k, z_{k+1}) \quad (5)$$

where d_B is a pre-specified domain distance between successive states. Euclidean distance after normalization, Mahalanobis distance, information-geometric distance, thermodynamic distance, or an empirically validated metric are all possible. If another measure predicts the domain outcome better under held-out testing, the framework permits that substitution.

Equation (4) is not intended to assign ordinary geometric length to arbitrarily fine noisy or stochastic trajectories. For Brownian-like paths, ordinary arc length can diverge as sampling resolution increases. The observation scale, filtering, event definition, or stochastic/information-geometric construction must therefore be declared inside M_B . Derivative-based geometry is admissible only when the trajectory is sufficiently resolved and finite at that scale; otherwise finite increments, transition counts, or stochastic measures should be used. A state-time result that changes solely because the same physical record was oversampled is not an admissible implementation.

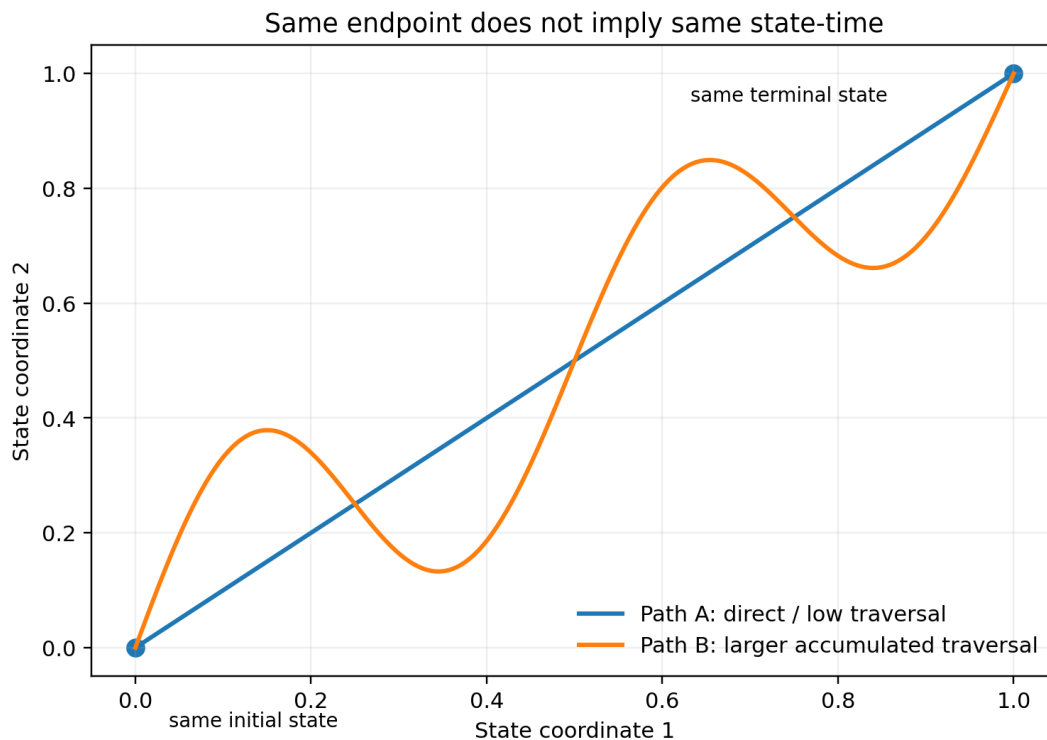


Figure 1. Two trajectories can share the same initial and terminal states while accumulating different state-time.

3.2. Energy drives change but does not define it

Energy is necessary to many physical state transitions, but energy input is not itself the proposed state-time coordinate. Equal energy inputs can produce different trajectories when the energy is delivered through different pathways, at different rates, under different environmental conditions, or when different transitions are accessible.

Represent the state dynamics generically as:

$$\dot{z}_B = F_B(z_B, \Phi_{E,in}, \Phi_{E,out}, \Phi_M, u, \eta, \mathcal{T}_B) \quad (6)$$

where $\Phi_{E,in}$ and $\Phi_{E,out}$ denote energy fluxes entering and leaving the chosen boundary, Φ_M matter fluxes where relevant, u control inputs, η environmental conditions, and $\mathcal{T}_B$ the set of accessible transitions. Equation (6) is deliberately generic. It expresses the central distinction: energy and environmental constraints drive, redirect, or remove capacity for change, while ξ_B^{full} measures the relevant trajectory actually taken under a chosen direct scalar rule. Net energy retained is therefore not sufficient to specify state evolution.

This separation avoids a common dimensional error. Joules measure energy; watts measure energy per time. A state-time metric should not be renamed energy. If two systems receive the same joules yet follow different normalized state trajectories, a given direct measurement rule may produce different ξ_B^{full} values.

3.3. Accessible transitions and the environment

The environment does more than add or remove energy. It can alter transition barriers, reaction pathways, phase behavior, heat-transfer coefficients, mechanical constraints, or control authority. Shielding gas in welding is an illustrative case: its central effect is not simply to reduce energy loss, but to change the chemical environment so that oxidation and contamination pathways become less accessible. Similarly, coolant in an engine exports thermal energy and changes which temperature-dependent failure transitions are reachable.

This motivates treating the accessible transition set $\mathcal{T}_B$ as an explicit part of the model. The same current state and nominal energy input need not yield the same next state when $\mathcal{T}_B$ differs.

3.4. Nested boundaries and hidden control cost

Boundary choice determines whether a transfer is classified as external or internal. If B_1 contains only an engine, heat transferred into coolant is an energy export. If B_2 contains engine plus cooling loop, the same heat transfer is internal redistribution and only radiator/environment exchange crosses the larger boundary. No physics changed; the accounting classification did. This reclassification motivates a formal requirement: boundary expansion may change the bookkeeping of exchanges, but it must not rewrite the already-defined state history of the embedded engine subsystem. Section 3.5 therefore treats the declared exchange/forcing record as part of Γ_B , so restriction can reclassify a larger-boundary internal transfer as a contained-boundary external exchange without changing the underlying physical event. Throughout this article, boundary expansion means analytical inclusion of an already-existing part of the same physical realization while retaining inherited resolution. Physically adding a controller, cooling device, material, or other subsystem changes the realized system and its trajectory; that is a system modification, not merely movement of the analytical boundary.

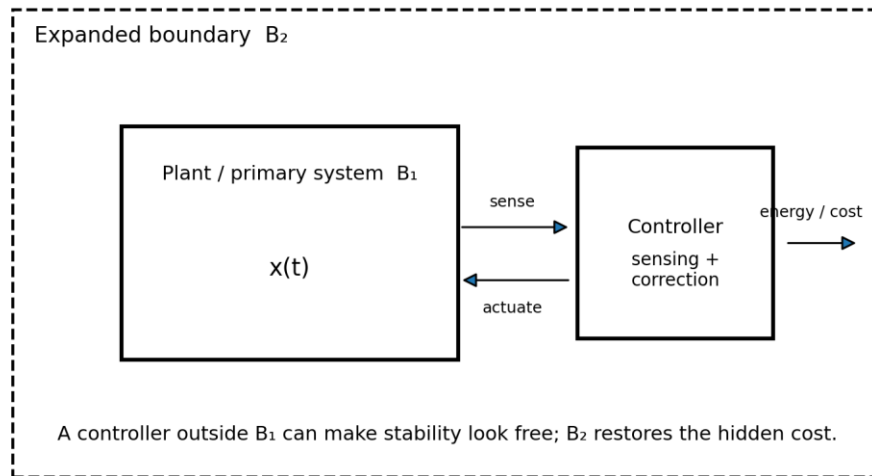


Figure 2. Expanding the boundary can reveal controller or stabilization costs that appear free under a narrow boundary.

The same logic applies to active control. A plant that appears stable only because an external controller continuously senses, computes, and actuates may have low internal variability under B_1 while imposing substantial state change and energy cost in the controller. Expanding the boundary to B_2 prevents the stability claim from hiding that burden.

This yields a practical accounting rule: whenever an optimization appears to eliminate state change locally, expand the boundary until the sensing, actuation, heat rejection, maintenance, or coupled subsystem that enables the result is either included or explicitly declared external. The next subsection formalizes what such expansion is allowed to change and what it must preserve.

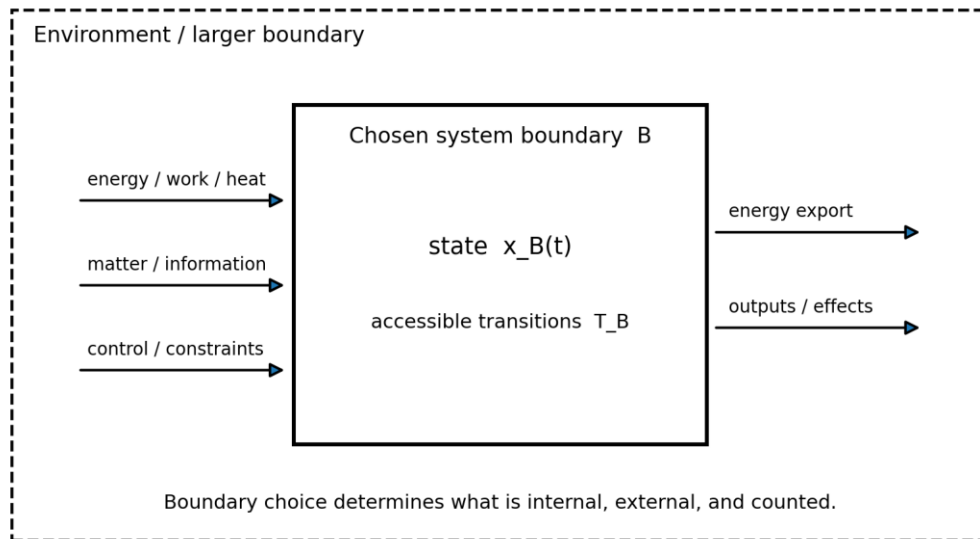


Figure 3. Boundary-first accounting: state evolution is interpreted only after the system and cross-boundary exchanges are specified.

3.5. Nested-boundary composition: restriction, coherence, and ledgers

The central boundary rule is an extension rule, not universal scalar additivity. Let $\mathcal{B}$ be a partially ordered collection of admissible system boundaries, ordered by inclusion. For $B_i \subset B_j$, let $\mathcal{H}(B)$ denote the admissible history records at boundary B . A record $\Gamma_B = (z_B, \Phi_B)$ contains internal state history plus the declared exchange/forcing records needed to interpret it and recover contained-boundary accounting. For a declared hierarchy and resolution, Φ_B is sufficiently resolved and labeled to support every admissible restriction. Let $\pi_{ji}: \mathcal{H}(B_j) \rightarrow \mathcal{H}(B_i)$ restrict a larger record to the inherited variables and resolution. Its state component recovers the inherited state history; its bookkeeping component re-expresses transfers crossing B_i as B_i boundary exchanges without changing the physical event. For an observed history Γ_{B_j} at inherited resolution:

$$\pi_{21}(\Gamma_{B_2}) = \Gamma_{B_1} \quad (7)$$

Equation (7) defines boundary expansion at inherited resolution: the larger description retains enough information to recover the prior subsystem history and relevant exchange accounting. A transfer internal to B_2 may reappear as external across B_1 ; the physical event, inherited state history, and measurement semantics are unchanged, only its boundary classification. Equation (7) alone is a consistency convention; the nontrivial requirement is coherent restriction across the hierarchy.

For $B_1 \subset B_2 \subset B_3$, the history restrictions must satisfy identity and composition laws:

$$\pi_{ii} = \text{id}, \quad \pi_{31} = \pi_{21} \circ \pi_{32} \quad (8)$$

Thus a whole-system description cannot produce one B_1 history when restricted directly and a different B_1 history when restricted through an intermediate boundary. If the history collections $\mathcal{H}(B)$ are treated as sets, the assignment $\mathcal{H}: \mathcal{B}^{\text{op}} \rightarrow \text{Set}$ together with the π maps is a contravariant functor, equivalently a Set-valued presheaf on the boundary poset. The present paper does not claim novelty for this mathematical pattern [11,12]. Its use here is to constrain how accumulated-history measurements may change when the analytical boundary changes.

Measurement rules obey an analogous restriction architecture. Let $\mathcal{M}(B)$ be the admissible measurement/accumulation rules at boundary B , and let $R_{ji}: \mathcal{M}(B_j) \rightarrow \mathcal{M}(B_i)$ restrict an expanded rule to the inherited subsystem semantics. Operationally, M_B is a declared rule bundle containing, as applicable, the observables or state coordinates to be evaluated, normalization/reference scales, observation resolution and sampling conventions, parameters or weights, event/edge/hysteresis conventions, and the evaluator/composition rule that maps an admissible history record to the chosen accumulated object. R_{ji} restricts that declared bundle to the inherited semantics rather than being chosen retrospectively from outcomes. A valid extension requires:

$$R_{21}(M_{B_2}) = M_{B_1} \quad (9)$$

with identity and composition:

$$R_{ji} = \text{id}, \quad R_{31} = R_{21} \circ R_{32} \quad (10)$$

Equations (9)-(10) distinguish extension from replacement. Boundary enlargement may not silently change the scale, weighting, event definition, or resolution used for inherited history. A different B_1 rule is allowed only as an explicitly declared model replacement. If added context shows that the earlier rule was physically incomplete, recalibration is permitted but begins a revised lineage rather than ordinary boundary expansion. When the rule collections $\mathcal{M}(B)$ are sets, $\mathcal{H}$ and $\mathcal{M}$ may be represented as contravariant functors over the boundary poset; no sheaf/gluing theory is required here.

Definition (accumulated-history measurement lineage). Over a declared nested boundary hierarchy $\mathcal{B}$, a measurement lineage is a compatible family of measurement rules $\{M_B\}$ indexed by $B \in \mathcal{B}$ together with coherent restrictions R_{ji} such that $R_{ji}(M_{B_j}) = M_{B_i}$ for every inherited-resolution inclusion $B_i \subset B_j$, with the identity and composition laws of Equation (10). A realized history family $\{\Gamma_B\}$ is compatible with that lineage when $\pi_{ji}(\Gamma_{B_j}) = \Gamma_{B_i}$ and the history restrictions obey Equation (8). Multiple realized history families may therefore be instantiated within the same measurement lineage. “Same measurement lineage” refers to preserved measurement semantics; each observed realization must additionally satisfy compatible history restriction. A rule change that fails inherited restriction begins a new lineage.

Given the history and rule restrictions in Equations (7) and (9), evaluation consistency can be summarized by the commuting relation $A_{B_i}(\pi_{ji}\Gamma_{B_j}; R_{ji}M_{B_j}) = A_{B_i}(\Gamma_{B_i}; M_{B_i}) = \xi_{B_i}^{\text{full}}$ for $B_i \subset B_j$ at inherited resolution. This relation is not an additional independent axiom; it records the consequence that evaluating the restricted history under the restricted rule reproduces the already-defined contained-boundary quantity. It does not require $\xi_{B_j}^{\text{full}} = \xi_{B_i}^{\text{full}}$: the larger boundary may add state, coupling, and accumulated history while preserving the contained evaluation semantics.

The geometric realization gives a concrete compatibility condition without choosing an arbitrary fixed slice of the newly included subsystem. Suppose the expanded state space factors locally as $Z_{B_2} = Z_{B_1} \times Z_C$ with projection $p_{21}(Z_1, Z_C) = Z_1$. Let g_{B_1} and g_{B_2} be the subsystem and expanded metrics. For every added-subsystem state z_C and for all inherited tangent directions v_1, w_1 , require:

$$g_{B_2}|_{(z_1, z_C)}((v_1, 0), (w_1, 0)) = g_{B_1}|_{z_1}(v_1, w_1) \quad \text{for all } z_C \quad (11)$$

In block-matrix notation, Equation (11) is equivalently the fiberwise condition:

$$W_{11}(z_1, z_{C,t}) = W_{B_1}(z_1, t) \quad \text{for all } z_C \quad (12)$$

Some candidate W_{B_2} fields are inadmissible because the meaning of pure inherited B_1 motion would depend on the newly included subsystem state. Cross terms may exist and coupling may alter actual trajectories; what cannot change under ordinary boundary expansion is the value assigned by the inherited rule to a pure inherited direction. In projection-compatible coordinates this direction is $(v_1, 0)$, yielding Equations (11)-(12). In invariant terms, the larger bilinear form restricted to the declared inherited tangent subbundle must equal g_{B_1} . Non-product spaces may use a projection/submersion-compatible restriction; the product form is the practical representation used here.

For a larger composite boundary, the conservative accounting object is a ledger rather than an immediate scalar. Let $\mathcal{L}(B)$ denote the domain of labeled ledger objects at boundary B , with $\Xi_B \in \mathcal{L}(B)$. The component decomposition itself is part of the declared boundary model: each entry has an identified subsystem or explicitly modeled interface, a state/history record, and an associated measurement rule or native history representation. If C is newly included material, controller, environment, or other already-present subsystem and I is an explicitly modeled state-bearing interaction/interface when needed, define:

$$\Xi_{B_2} = (\Xi_{B_1}, \Xi_C, \Xi_I) \quad (13)$$

The labeled tuple represents component-preserving bookkeeping: boundary expansion appends declared accounting entries rather than overwriting them, without implying that the heterogeneous entries belong to a common vector space, group, or other algebraic structure. The ledger preserves component histories in their own declared measurement coordinates or native representations. An interaction/interface entry Ξ_I is included only when I has been explicitly declared as a state-bearing component with its own history rule; a cross term in a full-boundary metric does not by itself create an interaction ledger entry. Coupling may alter the component trajectories through the dynamics and may also contribute to ξ_B^{full} through full-metric cross terms, but neither effect is allowed to erase or retroactively redefine a preserved ledger component.

Nested-boundary ledger construction must also be coherent under canonical regrouping. Here the symbol $\cong$ refers to equality of the declared component identities and their preserved history values up to ordering or parenthesization; it does not license an analyst to repartition the larger system after the fact. Component-preserving coherence is therefore conditional on the declared system decomposition: the present framework does not supply a unique canonical partition for an arbitrary coupled physical whole. For $B_1 \subset B_2 \subset B_3$, expanding from B_1 to B_3 directly and expanding first to B_2 and then to B_3 must preserve an equivalent final ledger after the same declared components are identified:

$$\Xi_{B_3}(\text{direct}) \cong \Xi_{B_3}(\text{staged}) \quad (14)$$

This is a coherence requirement on nested-boundary accounting. It does not claim a general gluing theorem for overlapping non-nested subsystems. Such gluing would require additional compatibility conditions on overlaps and is outside the present scope.

A scalar ledger summary is secondary to any direct full-boundary metric. Let $\mathcal{S}(B)$ denote admissible declared scalarization rules, with each $S_B: \mathcal{L}(B) \rightarrow \mathbb{R}_{\geq 0}$ where aggregation is meaningful; no scalarization is required for incommensurate entries. Scalarizations are boundary-local by default. Cross-boundary comparison requires a coherent restriction family $T_{ji}: \mathcal{S}(B_j) \rightarrow \mathcal{S}(B_i)$, with identity/composition laws and $T_{ji}(S_{B_j}) = S_{B_i}$. This preserves inherited weights, normalization, and aggregation meaning without embedding a smaller ledger into a larger one or assuming neutral values for new components. It does not require $\sigma_{B_j} = \sigma_{B_i}$. If no such restriction exists, σ values remain boundary-local and are not directly comparable. For a compatible nonnegative linear scalarization:

$$\sigma_{B_2} = S_{B_2}(\Xi_{B_2}) = w_1 \xi_{B_1}^{\text{full}} + w_C \xi_C^{\text{full}} + w_I \xi_I^{\text{full}}, \quad w_j \geq 0 \quad (15)$$

For Equation (15), if $S_{B_1}(\Xi_{B_1}) = w_1 \xi_{B_1}^{\text{full}}$, then $T_{21}(S_{B_2}) = S_{B_1}$, simply retains the inherited term and weight w_1 while omitting newly added C and I terms from the restricted rule. For a nonlinear coupled S_{B_2} , a valid T_{21} need not exist; in that case cross-boundary σ comparison is inadmissible. Under a common compatible linear rule calibration and independent subsystems, $\xi_I = 0$ and the special additive case follows:

$$\sigma_{B_2} = \xi_{B_1}^{\text{full}} + \xi_C^{\text{full}} \quad (16)$$

Universal monotonicity is not asserted across incompatible domain units. The invariant is component preservation plus coherent restriction of the relevant semantics. When a compatible nonnegative ledger scalarization exists, σ_B is monotone under addition of nonnegative components only within the declared aggregation rule. If cross-boundary scalar comparison is claimed, T_{ji} must preserve inherited component weights, normalizations, and aggregation meaning; changing those semantics begins a new aggregation-rule lineage rather than an inherited extension. No equality is generally required between σ_B and ξ_B^{full} . The former aggregates declared component histories; the latter is produced by a rule on the complete coupled state and may contain information, such as cross-coupled traversal, that cannot be reconstructed from component scalar histories alone.

Finally, every direct scalar state-time ξ_B^{full} is required to be temporally additive within a fixed boundary and measurement rule:

$$\xi_B^{\text{full}}[t_0, t_2] = \xi_B^{\text{full}}[t_0, t_1] + \xi_B^{\text{full}}[t_1, t_2] \quad (17)$$

Together, nonnegativity and temporal additivity of direct scalar state-time rules, contravariant restriction coherence, inherited rule calibration, fiberwise geometric compatibility, coordinate consistency, declared component identity, and component-preserving ledger composition restrict admissible state-history constructions beyond the generic instruction to integrate a useful predictor. Non-additive native history objects may remain in the ledger under their own declared composition laws without being mislabeled as ξ_B^{full} .

3.6. Boundary expansion is not coarse-graining

Boundary expansion and model abstraction are different operations. Boundary expansion, as used in Equations (7)-(14), enlarges what is included while retaining the previously specified subsystem description. Coarse-graining deliberately reduces resolution, merges states, averages variables, or otherwise discards information. A useful coarse-grained model may therefore be unable to reconstruct the detailed Γ_B , exactly; that information loss is not itself a violation of nested-boundary composition.

The distinction must be declared. If c_B maps a detailed history Γ_B to a reduced history $\bar{\Gamma}_B$, then c_B is a coarse-graining map, not a boundary projection. Exact projection consistency is mandatory only for expansion at inherited resolution. When an analysis claims consistency across both expansion and coarse-graining, the two operations may be tested for commutation at the retained resolution:

$$c_{B_1} \circ \pi_{21} \approx \bar{\pi}_{21} \circ c_{B_2} \quad (18)$$

where $\approx$ denotes equality within the declared coarse-graining tolerance or equivalence class. Failure of Equation (18) need not invalidate either model, but it means the coarse-grained and expanded descriptions are not interchangeable for the claimed history measure.

3.7. Functional equivalence and history equivalence

Two realizations may produce the same prescribed outputs at a chosen boundary and therefore be interchangeable for a functional question while accumulating different internal histories. A controller can hold plant output nearly fixed while its own computation, switching, heat rejection, or actuator history grows. Likewise, a larger model can be functionally black-box equivalent at its external ports while differing internally. State-time is relevant only when

that internal history matters to the question, when the boundary is expanded to include the burden, or when the accumulated history predicts a later outcome.

3.8. Worked nested-boundary example

A synthetic example shows that the framework rules distinguish admissible from inadmissible extensions before any outcome fitting and also illustrates why a direct full-boundary scalar must be distinguished from a ledger scalarization. Let external time be $t \in [0,2]$. Begin with a one-state plant boundary B_1 whose state x follows the triangular history $x: 0 \rightarrow 1 \rightarrow 0$ at unit speed. Choose the geometric rule $W_{B_1} = [1]$. The accumulated plant history is therefore $\xi_{B_1}^{\text{full}} = \int |\dot{x}| dt = 2$.

$$B_1: z_1 = x, \quad W_{B_1} = [1], \quad \xi_{B_1}^{\text{full}} = 2$$

Expand the analytical boundary to $B_2 = B_1 + C$ by including an already-present one-state controller c with history $c: 0 \rightarrow 0.5 \rightarrow 0$. Let the controller component use $W_C = [1]$, so $\xi_C^{\text{full}} = 1$. Define an expanded geometric metric with $\rho = 1/2$:

$$W_{B_2} = [[1, 1/2], [1/2, 1]]$$

This extension is admissible and positive definite: its eigenvalues are 0.5 and 1.5. The inherited block is $W_{11} = 1$ for every controller state, so a pure inherited plant direction keeps exactly its B_1 calibration. Cross-coupling is still allowed: for simultaneous velocities $(\dot{x}, \dot{c})$ the full B_2 traversal contains the $2\rho\dot{x}\dot{c}$ term, but that coupling does not redefine the numerical meaning of plant-only motion. The history restriction is $\pi_{21}(x,c)=x$ and the rule restriction $R_{21}(W_{B_2})=W_{B_1}$. The conservative component ledger is $\Xi_{B_2} = (2,1)$ before any optional scalarization.

The direct full-boundary geometric measure is nevertheless not 3. During each half of the trajectory, $|\dot{x}|=1$ and $|\dot{c}|=0.5$ with the same sign, so Equation (4) gives $\gamma_{B_2}^{\text{full}} = \sqrt{(1 + 0.25 + 0.5)} = \sqrt{1.75}$. Therefore:

$$\xi_{B_2}^{\text{full}} = 2\sqrt{1.75} \approx 2.646$$

If the two ledger components were independently judged commensurate and an equal-weight additive summary were chosen, then $\sigma_{B_2} = S_{B_2}(\Xi_{B_2}) = 2 + 1 = 3$. The values 2.646 and 3 are not competing definitions of one quantity. $\xi_{B_2}^{\text{full}}$ measures traversal of the complete coupled (x,c) state under W_{B_2} , including its cross term; σ_{B_2} summarizes the preserved component histories under a separate aggregation rule. In general $\xi_B^{\text{full}} \neq \sigma_B$, and neither quantity is required to reconstruct the other. This σ_{B_2} is a boundary-local summary unless its aggregation rule is explicitly shown to extend the corresponding smaller-boundary scalarization semantics.

Contrast this with a proposed expanded metric that makes the inherited block depend on the newly included controller state:

$$W'_{B_2}(x,c) = [[1 + c^2, 0], [0, 1]]$$

This is inadmissible as ordinary boundary expansion. At $c = 0.5$, a unit pure- x direction has squared norm 1.25 rather than 1, so the meaning of the previously calibrated B_1 history has changed merely because C was added. Such a rule could be studied only as an explicit replacement/recalibration of the B_1 model, not as an inherited extension.

Now expand the analytical boundary again to $B_3 = B_2 + T$ by including an already-present one-state thermal-management subsystem θ with history $\theta: 0 \rightarrow 0.25 \rightarrow 0$ and $W_T = [1]$, giving $\xi_T^{\text{full}} = 0.5$. Choose:

$$W_{B_3} = [[1, 1/2, 0], [1/2, 1, 1/4], [0, 1/4, 1]]$$

The upper-left 2×2 block is exactly W_{B_2} , and W_{B_3} is positive definite (its eigenvalues are 1 and $1 \pm \sqrt{5}/4$, all positive). Thus $R_{32}(W_{B_3})=W_{B_2}$; restricting again gives $R_{21}R_{32}(W_{B_3})=W_{B_1}$. Direct restriction gives the same result, $R_{31}(W_{B_3})=W_{B_1}$. Likewise, $\pi_{32}(x,c,\theta)=(x,c)$, $\pi_{21}(x,c)=x$, and $\pi_{31}(x,c,\theta)=x$, so $\pi_{31}=\pi_{21} \circ \pi_{32}$. The declared component

decomposition is plant/controller/thermal-management, with no separate interface state in this toy model. The ledger obtained directly or in stages is therefore the same up to ordering and parenthesization of those declared components:

$$\Xi_{B_3}(\text{direct}) = (2, 1, 0.5) \cong \Xi_{B_3}(\text{staged})$$

This toy construction is intentionally simple. Its purpose is not to establish a preferred physical metric, but to demonstrate non-emptiness of the admissible class, the practical difference between extension and replacement, and the semantic distinction between a direct full-boundary measure and an optional ledger scalarization. The framework constrains how a larger model may inherit previously calibrated history semantics even though the larger system may contain new states, new couplings, and new component histories.

3.9. Information-geometric instantiation

The same architecture can be instantiated with an information-geometric measurement rule rather than an engineering weighting matrix. Let $q_B(t)$ parameterize a family of statistical states $p_B(y|q_B)$, and let M_B use the Fisher-information metric $I_B(q_B)$. A direct full-boundary information-length history is then:

$$\xi_B^{\text{full}} = \int \sqrt{\dot{q}_B^T I_B(q_B) \dot{q}_B} dt$$

For a nested expansion $B_1 \subset B_2$ with local coordinates $q_{B_2}=(q_1, q_C)$, one admissible inherited-resolution extension of the B_1 information-length calibration requires the inherited block to satisfy $[I_{B_2}]_{11}(q_1, q_C)=I_{B_1}(q_1)$ throughout the relevant q_C fiber. This is an admissibility assumption of the present framework, not a claim that Fisher information generically behaves this way whenever a statistical model is enlarged. Added nuisance parameters can change the effective information available about parameters of interest, for example through projection or Schur-complement constructions [13]. Cross blocks may remain nonzero and can therefore change $\xi_{B_2}^{\text{full}}$ by encoding coupled statistical motion. The component ledger still preserves the previously calibrated B_1 history separately, and any σ_{B_2} formed from ledger entries is a distinct aggregation. If statistical-model enlargement genuinely changes the effective inherited information, the framework classifies that change as a new calibration/model lineage rather than silently treating it as unchanged boundary expansion. This connects the framework to information-length and information-geometric constructions used in nonequilibrium and stochastic systems [6,7], while making clear that Equation (12)-type block preservation is a selected extension rule rather than a universal fact of inference geometry.

3.10. Non-geometric restriction: threshold-event counting

The restriction architecture is not limited to Riemannian or weighted-path metrics. Consider a plant boundary B_1 with a monitored temperature $T(t)$. Let M_{B_1} count upward crossings of a declared threshold T^* using a fixed hysteresis band, sampling rule, and interval-edge convention, producing the accumulated event history N_{B_1} . Because hysteresis has memory, the event detector also carries a logical state $h(t) \in \{\text{armed, disarmed}\}$. The threshold, hysteresis band, sampling rule, edge convention, and initial detector state are part of the calibrated event semantics. Expand to B_2 by including a controller with its own switching-event count N_C . The rule restriction R_{21} acts on the rule specification - threshold, hysteresis, sampling, event convention, and detector-state semantics - and must recover M_{B_1} exactly. Separately, the ledger projection removes the realized controller-history component from $\Xi_{B_2} = (N_{B_1}, N_C)$ and recovers the realized plant history N_{B_1} . No metric tensor or state-space arc length is required.

By contrast, an expanded rule that changes the inherited plant event to 'count a T^* crossing only when the controller is high,' changes the hysteresis band, resets detector state without declaration, or changes the sampling resolution without declaring coarse-graining does not restrict to M_{B_1} . It is therefore a new measurement model rather than an inherited extension. For a temporal partition at t_1 , the terminal detector state $h(t_1)$ from the first interval is inherited as the initial detector state for the second interval, and the same interval-edge convention is retained. With that state carried forward, $N_{B_1}[t_0, t_2] = N_{B_1}[t_0, t_1] + N_{B_1}[t_1, t_2]$ without double counting. This small example gives an operational

meaning to $R_{21}(M_{B_2})=M_{B_1}$, for a genuinely non-geometric measure and narrows the role of Equations (11)-(12) to the geometric subclass rather than the framework as a whole.

4. Worked Examples

4.1. Finite transformation: cooking a steak

Let the steak be the chosen boundary. A coarse state vector might include center temperature, surface temperature, water content, protein-denaturation markers, and surface-reaction state. For a teaching example, suppose a pre-specified scalar rule assigns one reference unit of accumulated history to a chosen raw-to-target-cooked trajectory.

If a high-temperature oven drives that reference accumulated transformation in one hour, its average accumulation rate is one reference unit per hour. If a lower-temperature oven produces the same calibrated accumulated quantity in two hours, its average rate is one-half reference unit per hour. The purpose of the example is not to assert that real steaks follow identical trajectories. Different heating schedules can produce different gradients, moisture loss, surface chemistry, and structural change even when center temperature matches.

The example separates three concepts that are often conflated: elapsed time, terminal measurement, and accumulated transformation. The steak also demonstrates nested boundaries: the surface and center can experience very different state-time rates while remaining parts of the same physical object.

4.2. Continuous operation: an LED under stable and rippled drive

Consider two otherwise matched LED assemblies operated at the same average electrical power. Drive A produces nearly steady current and junction temperature after startup. Drive B contains a 100 Hz ripple of sufficient magnitude to create repeated excursions in current, junction temperature, or stress-sensitive internal variables. This is a waveform comparison only; it is not a claim that constant-voltage drive is generally superior to constant-current drive.

Under a direct full-boundary metric that weights those variables, the rippled system traverses additional state-space distance on every rise and fall. Equal average power and equal elapsed time therefore need not imply equal ξ_B^{full} under that chosen calibration. The framework therefore motivates the empirical hypothesis that, if repeated excursions contribute to wear, a well-calibrated direct history measure may correlate with degradation better than hours-on alone.

This example is especially useful because there is no natural finished state. The device's useful function is continuous. State-time becomes an accumulated operating-history variable rather than progress toward a terminal goal.

4.3. Boundary expansion: engine and cooling system

Let B_1 contain only the engine. Combustion supplies energy; heat transfer to coolant is an export across the boundary. Under inadequate cooling, more thermal energy remains in engine components, which changes the accessible transition landscape toward lubricant breakdown, deformation, knock, or seizure. Under effective cooling, the same engine output can correspond to a different internal trajectory.

Now expand the boundary to $B_2 = \text{engine} + \text{coolant loop} + \text{radiator}$. Heat transfer from engine metal to coolant is no longer a boundary loss; it is an internal exchange. The radiator/environment interface becomes the relevant thermal boundary. This does not make one boundary more true than the other. Each answers a different question. The framework requires the boundary to be explicit so state-time and energy accounting are not interpreted outside their scope.

4.4. Real-system boundary check: Luminus LED boundary reclassification and thermal coupling

A real-system boundary check can be constructed from public Luminus engineering documentation [14-16]. The Generation 4 CXM-14 datasheet lists the nominal 36 V CXM-14-35-80-36-AC40-F5-3 (3500 K, 80 CRI) at a 720 mA test current and $T_j = 85^\circ\text{C}$, with 3,840 lm typical output at that declared operating condition [14]. The same

datasheet states that its COB products are measured at fixture-relevant operating temperature, while 25 °C flux values are calculated for reference [14]. The significance for the present framework is not the numerical flux itself, but the explicitly specified operating/measurement context: the reported quantity is attached to a declared operating state rather than detached from the conditions that give it meaning.

Independent horticultural-lighting literature makes the package-to-fixture boundary issue explicit. Kusuma et al. identify four fixture-level factors beyond LED-package efficacy - drive current, junction temperature, driver efficiency, and optical losses - and note that LED junction temperature in fixtures is typically around 85 °C [17]. Their representative typical-parameter calculation yields fixture efficacy equal to 67% of the reported LED-package efficacy after these factors are applied [17]. The same review cautions that package-based LM-80 lifetime projections do not include fixture optical-loss mechanisms, accelerated aging caused by higher operating temperatures, or catastrophic LED-driver failure [17]. These observations do not validate the present framework, but they independently show why package, fixture, and environment boundaries answer different performance and reliability questions.

Luminus thermal-design documentation then makes the larger boundary explicit. A single-device thermal model relates junction temperature to ambient temperature, thermal resistance, and dissipated power, and refines dissipated power to $P_{\text{electrical}} - P_{\text{optical}}$ when optical output is accounted for [15]. More directly, a separate Luminus application note for the SBM-40 multi-die package publishes a manufacturer-characterized thermal cross-talk matrix in which each die's junction rise depends on both its own dissipated power and the dissipated power of neighboring dies [16]. The CXM-14 and SBM-40 are not treated as identical hardware or as one literal experimental $B_1 \subset B_2$ sequence: the CXM-14 demonstrates explicit single packaged COB specification context, while the SBM-40 supplies a public coupled-package realization of the boundary-reclassification problem.

For the SBM-40-SC 'real' matrix, Luminus gives off-diagonal thermal coefficients in addition to each die's self-heating coefficient [16]. In the manufacturer's worked case, optical-subtracted dissipated powers are (1.9, 2.7, 0.6, 0.8) W for the red, green, blue, and white dies, respectively. Applying the published matrix gives junction-to-case rises of approximately (13.8, 24.0, 10.5, 11.1) °C, consistent with the application note [16]. Table 1 separates each physical total into the die's own diagonal/self contribution and the thermal influence of the other dies.

Die	Self contribution	Other-die contribution	Full ΔT_{j-c}	Other-die share
Red	7.79 °C	5.98 °C	13.77 °C	43.4%
Green	19.58 °C	4.41 °C	23.98 °C	18.4%
Blue	2.83 °C	7.70 °C	10.53 °C	73.1%
White	4.48 °C	6.66 °C	11.14 °C	59.8%

Table 1. Decomposition of the Luminus SBM-40-SC manufacturer example into each die's self contribution and neighboring-die thermal influence, calculated from the published real thermal matrix and optical-subtracted powers [16]. For a one-die boundary, the neighboring contribution is external; for the full package boundary, the same contribution is internal coupling.

The coupled burden is substantial, but boundary expansion does not change the physical junction-temperature response. For the red die, the total modeled junction-to-case rise is 13.77 °C at either boundary. Under B_1 = red die, that total may be accounted as 7.79 °C of internal self-heating plus 5.98 °C of external cross-boundary thermal influence from the neighboring dies. Under B_2 = four-die package, the same 13.77 °C is accounted as 7.79 °C of red-die self-heating plus 5.98 °C of internal neighboring-die coupling. For the blue die, the corresponding decomposition is 2.83 °C self plus 7.70 °C neighboring influence, totaling 10.53 °C. Thus expansion changes attribution and resolution, not the underlying physical response: a contribution that was external at B_1 becomes internal at B_2 while the inherited self-heating coefficient and total physical response remain unchanged.

This is a static physical analogue of the boundary-reclassification rule, not an end-to-end accumulated-history validation. The published matrix identifies self and coupled contributions within one characterized model; it does not document a historical B_1 -to- B_2 calibration sequence or show predictive superiority over an established LED reliability model. Its relevance is narrower: the same thermal burden changes classification as the analytical boundary moves while the inherited local term remains interpretable. Luminus also notes that MCPCB layout,

thermistor location, and solder choice can alter case-to-reference thermal resistance [16], pointing to a natural next boundary in which board, reference, and environment state must be declared. A prospective state-time test would require time-resolved electrical, thermal, and optical trajectories and a pre-specified measurement rule.

5. Structural Consequences and Empirical Hypotheses

Three claim types must remain separate: consequences of the definitions, conformance tests for proposed implementations, and empirical hypotheses. A failed conformance test means the construction does not instantiate this methodology; empirical failure begins when a pre-specified implementation is asked to explain or predict an outcome.

- Structural consequence - Matched external duration does not imply matched direct full-boundary state-time. Systems exposed for the same laboratory time but subjected to different forcing histories can produce different ξ_B^{full} when the chosen history rule is history-sensitive.
- Structural consequence - Matched terminal state does not imply matched direct full-boundary state-time. Two trajectories that converge to the same measured endpoint can accumulate different ξ_B^{full} under a history-sensitive rule.
- Structural consequence - Matched total energy or matched average power does not imply matched direct full-boundary state-time. Waveform, environmental constraints, and accessible transitions can change the path even when integrated energy is similar.
- Structural consequence - Boundary expansion can change attribution and therefore apparent efficiency or stability rankings when a narrow boundary excludes controller, cooling, maintenance, or coupled-system burden that becomes internal or separately visible under a larger boundary.
- Empirical hypothesis - A pre-specified, calibrated direct history measure ξ_B^{full} should improve prediction of at least one domain outcome - such as wear, degradation, transformation, or control burden - beyond elapsed time, conventional state endpoints, total energy, and the strongest relevant established domain metric. A ledger-derived σ_B may also be tested if its aggregation rule is fixed independently and, when cross-boundary comparison is claimed, uses an aggregation rule that restricts coherently to the inherited semantics. Failure to add value falsifies the predictive claim for that implementation and domain; it does not by itself prove that no useful boundary-relative history measure exists in any other system.

Empirical testing rule - Metric or scalarization substitution should not rescue the theory post hoc. If predictive performance appears only after outcome-specific state variables, weights, or aggregation rules are chosen after seeing the result, the test has failed. A useful implementation should survive held-out data and clearly state when it reduces to an existing metric.

Structural admissibility test - Boundary expansion provides tests independent of outcome fitting. If B_1 is measured first and the experiment is later instrumented as $B_2 \supset B_1$ at inherited resolution, restriction of Γ_{B_2} must recover Γ_{B_1} . For a chain $B_1 \subset B_2 \subset B_3$, direct and staged restrictions must agree, and the expanded measurement rule must restrict to the inherited rule. If ledger scalarizations are compared across boundaries, the aggregation-rule restrictions T must also return the inherited scalarization semantics. In a geometric product-space implementation, the expanded metric must satisfy the fiberwise inherited-direction conditions in Equations (11)-(12). A formulation that only achieves consistency by recalibrating B_1 after expansion is not an admissible implementation of the framework.

6. Suggested Experimental Program

6.1. Minimal physical test

These experiments are deliberately prospective. Freezing the framework before outcomes are observed prevents later testing from redefining it to fit the data. Start with a system that has clean sensing, repeatable forcing, and a

known degradation or transformation endpoint. The first question is whether boundary-first, history-sensitive accounting adds predictive value beyond simpler descriptions and the strongest established domain metric.

- Choose a narrow, explicit boundary B and identify a small state vector x_B .
- Normalize the state variables using engineering-relevant scales and specify the candidate measurement rule M_B - including any W_B if a geometric implementation is used - before testing outcomes.
- Create multiple forcing histories with matched laboratory duration and, where possible, matched total energy or matched terminal state.
- Compute ξ_B^{full} for each run when a direct scalar rule is being tested; if a ledger scalarization σ_B is also proposed, compute it separately under its pre-specified aggregation rule. When σ_B values are compared across nested boundaries, pre-specify and verify the aggregation-rule restriction $T_{ji}(S_{B_j}) = S_{B_i}$ so inherited weights, normalizations, and aggregation meaning remain unchanged. Compare each claimed predictor with conventional baselines such as elapsed time, peak temperature, RMS current, energy consumption, and the strongest relevant established damage or transformation metric.
- Separate predictor variables from the held-out outcome whenever predictive superiority is claimed. A degradation, optical, thermal, or other observable used directly inside M_B should not simultaneously serve as the nominally unseen outcome unless the analysis is explicitly labeled descriptive/explanatory rather than predictive.
- Test whether the pre-specified ξ_B^{full} and/or σ_B predicts the chosen outcome on held-out runs. If a simpler existing metric performs equivalently, report the reduction rather than renaming it. Refit or expand a measurement or scalarization rule only in a new training cycle and re-test on fresh held-out data.

6.2. Nested-boundary coherence test

Instrument three nested descriptions of the same physical apparatus, for example $B_1 = \text{plant}$, $B_2 = \text{plant} + \text{controller}$, and $B_3 = \text{plant} + \text{controller} + \text{thermal-management subsystem}$. Calibrate M_{B_1} first. Construct M_{B_2} and M_{B_3} without changing the inherited B_1 rule. Verify (i) $\pi_{31}\Gamma_{B_3}$ and $\pi_{21}\pi_{32}\Gamma_{B_3}$ recover the same B_1 history, (ii) R_{31} and $R_{21}R_{32}$ return the same inherited measurement rule, and (iii) direct and staged ledger expansion produce equivalent component histories. For a geometric product-space realization, vary the newly included subsystem state while holding the inherited state direction fixed and test whether $W_{11}(z_1, z_{C,1})$ remains equal to the previously calibrated $W_{B_1}(z_1, t)$ across the relevant fiber. This is a structural validation of the implementation before any combined scalar is attempted.

6.3. Expansion-versus-coarse-graining test

Repeat one nested-boundary analysis after deliberately coarse-graining the recorded state, for example by reducing temporal resolution or aggregating several state variables. Declare the coarse-graining map and tolerance before comparison. Determine whether the reduced description preserves the claimed subsystem history at that resolution and whether coarse-graining before versus after boundary projection yields equivalent retained histories. A failure here is informative: it identifies a resolution at which the proposed state-time quantity is not preserved and prevents coarse-graining from being mistaken for ordinary boundary expansion.

6.4. LED example

For an LED test bed, candidate state observables could include drive current, forward voltage, junction-temperature proxy, board temperature, driver ripple, and thermal-management state. Optical flux or spectral shift may instead be reserved as held-out degradation outcomes when predictive superiority is claimed; if either is included inside M_B , then it cannot simultaneously serve as an independent target. The Luminus case in Section 4.4 is intentionally a boundary/coherence demonstration rather than a lifetime-validation study: the published matrix provides a physically grounded coupled-state calculation but not the time-resolved degradation trajectories needed to pre-specify and test a full accumulated-history predictor. A prospective experiment should therefore log the electrical and thermal trajectories continuously, declare the drive architecture and boundary decomposition in

advance, calibrate M_B before inspecting degradation, and compare ξ_B^{full} and any declared σ_B against elapsed time and the strongest established reliability model for the chosen devices.

6.5. Computational test

A computational system offers a different advantage: internal states can be logged at high temporal resolution without destructive sensing. A recurrent or looped model can be driven through repeated state updates while hidden-state distance, output change, energy consumption, and convergence behavior are measured. If state-time is merely redundant with iteration count or energy, the distinction should fail quickly. If not, computational systems may provide a rapid way to refine the metric before translating it back to physical systems.

7. Relation to Existing Work

Boundary-relative state-time is closest in spirit to several established families of ideas but is not identical to any one of them. Endochronic theory is a particularly important precedent: Valanis introduced an intrinsic time not measured by a clock and, in early formulations, tied that intrinsic time to path length in strain space [18,19]. Material-time research in glasses likewise treats physical aging as governed by an internal clock and has recently linked material-time increments to distance traveled in configuration space [2-4]. These precedents make clear that intrinsic-time and path-length constructions are not new by themselves.

Thermodynamic length provides a metric geometry on thermodynamic-state space and is related to dissipation and optimal finite-time transformations [5]. Information length similarly accumulates distinguishable changes in probability distributions and explicitly depends on trajectory history [6,7]. Dynamical activity in stochastic thermodynamics counts or weights transitions and has become central to kinetic uncertainty relations, response bounds, and thermodynamic speed limits [8-10]. Engineering and biomedical practice also contains many accumulated-dose constructions; the CEM43 thermal-dose formulation, for example, maps a temperature history into a cumulative equivalent exposure [20]. These works demonstrate that integrating a state-dependent or path-dependent rate is an established modeling pattern.

Boundary and interconnection modeling also has extensive precedent. Port-Hamiltonian formulations treat stored energy and boundary power exchange as first-class objects [21]; life-cycle assessment makes boundary choice and expansion explicit [22]; Willems centers admissible trajectories and interconnection [23]; compositional open-systems theory formalizes assembly through interfaces [24,25]; and coarse-graining open Markov processes separates composition from information-reducing abstraction [26]. The Set-valued history and rule assignments used here form presheaves on the boundary poset [11,12]. These structures are precedents, not contributions claimed here.

Applied sheaf theory provides a closer precedent for local-to-global measurement consistency. Robinson's sensor-integration framework represents heterogeneous local observations with restriction maps, tests consistency across overlapping local descriptions, and emphasizes faithful integration of local measurement meaning before or alongside fusion [27]. That work substantially anticipates the use of restriction structure to preserve heterogeneous measurement semantics. It does not, however, address the particular object studied here: accumulated histories indexed by nested analytical boundary expansion, reclassification of exchanges as the boundary moves, extension-versus-replacement of accumulation rules, the distinction between direct full-boundary history and a component-preserving ledger, or coherent restriction of optional aggregation rules. The present proposal therefore makes no novelty claim for sheaf or local-to-global consistency machinery itself. In this paper, measurement lineage is also distinct from ordinary data lineage or provenance, which tracks the origin, transformation, and derivation of data objects; here lineage denotes preservation of measurement semantics under analytical boundary inclusion [28].

Measurement science provides another adjacent lineage. Measurement-invariance research asks whether a construct retains comparable meaning across groups or occasions [29], while metrological traceability links a result to a specified reference through a documented calibration chain [30]. Neither directly addresses nested-boundary accumulated histories. Here, 'rule calibration' is broader than VIM metrological calibration: it denotes the declared parameterization and reference semantics of M_B unless traceability is separately established. These literatures

nevertheless provide precedents for preserving measurement meaning and declaring new lineages when that meaning changes; the present proposal applies a related discipline to boundary-indexed histories, exchange reclassification, and coherent restriction.

Recent work in organisational cybernetics likewise treats system persistence as temporally structured. Osejo-Bucheli's Temporal Viability Theory defines viability as a finite, recursively accumulated timespan shaped by growth, decay, intervention, latency, and recursive temporal organisation [31]. That framework asks how temporal structure conditions the persistence and coherence of viable socio-technical systems. Boundary-relative state-time addresses a different object: accumulated state evolution and preservation of measurement semantics as analytical boundaries change. Viability duration and accumulated state history therefore need not coincide; two systems may remain viable for the same elapsed interval while accumulating different boundary-relative histories, or accumulate similar histories while differing in remaining viability.

A third close mathematical precedent is the classical theory of additive functionals of stochastic and Markov processes. Dynkin explicitly described an additive functional as a way to measure time while accounting for the development of the process [32], and later treatments develop positive continuous additive functionals together with time-change theory for Markov processes [33]. The present framework therefore does not claim novelty for temporally additive trajectory clocks, process-dependent time changes, or the additivity law used by ξ_B^{full} . Its proposed contribution begins with the additional boundary-indexed lineage constraints: inherited history and rule restriction, exchange reclassification, component-preserving ledgers, and coherent aggregation-rule restriction across nested analytical boundaries.

Projective systems provide another direct precedent. Vilela Mendes [34] emphasizes bonding-map composition and consistent measures across ordered families of physical descriptions. Projective consistency is therefore established mathematics. Here the indexed objects are different: accumulated system histories, measurement-rule semantics, component ledgers, and optional scalarizations under analytical boundary inclusion rather than probability measures across scales.

Formal methods provide another close structural precedent. Compositional refinement preserves designated behavior under refinement and composition [35], while conservative-extension results require richer theories to leave original behavior or derivability unchanged [36]. That is close to the extension-versus-replacement distinction used here. Conservative extension and compositional refinement are not claimed as new; the specific proposal is to apply that preservation discipline to boundary-indexed accumulated histories, measurement semantics, exchange reclassification, component ledgers, and optional aggregation-rule lineage.

The proposed synthesis is therefore not a new theory of presheaves, projective limits, metrics, or additive functionals. Its primary claim is a systems-measurement discipline for accumulated histories: across nested analytical boundaries, previously declared histories and measurement semantics must remain recoverable and coherent if the larger description is to count as an extension of the same lineage. Geometric compatibility, component-preserving ledgers, and coherent aggregation-rule restriction operationalize that claim for particular realizations. A direct full-boundary scalar ξ_B^{full} may encode coupled traversal, whereas σ_B summarizes declared ledger components under a separate aggregation rule. The present scope is nested boundary hierarchies; gluing over overlapping non-nested subsystems remains outside scope.

8. Limitations and Falsifiability

The framework does not specify a universal measurement rule M_B , direct scalar ξ_B^{full} , or ledger aggregation S_B . Domains differ and ledger components need not be commensurate; scalarizations are boundary-local unless aggregation restrictions exist. Direct ξ_B^{full} quantities are limited to additive coordinates, while non-additive native histories may remain in Ξ_B . If newly modeled context shows an inherited rule was physically incomplete, the

response is explicit model replacement and a new lineage, not silent recalibration. The Fisher-information example illustrates this case; the Luminus case illustrates reclassification with an inherited local term preserved.

The framework can be made vacuous if state variables, boundaries, weights, sampling scales, component decompositions, or scalarization rules are chosen after observing outcomes. Pre-specification, cross-validation, and comparison with simpler baselines are therefore essential. A pre-specified ξ_B^{full} or σ_B that adds no value beyond the strongest established metric fails for that use. Structural admissibility instead asks whether the declared restriction, rule, coordinate, sampling, component, ledger, and aggregation-lineage requirements are satisfied. This is methodological conformance, not falsification of a law of nature. Repeated failure to produce nontrivial useful implementations would nevertheless weaken the framework's scientific value.

State-time is not proper time in relativity, a replacement for thermodynamic time, or a claim that physical systems literally experience subjective time. The term time is used because ξ_B^{full} , when a scalar rule is chosen, parametrizes accumulated internal evolution against external laboratory time. More conservatively, boundary-relative state-time names a class of accumulated-history constructions that obey the proposed boundary and coherence rules; members of that class need not share a common physical dimension.

Coordinate and sampling dependence remain limitations. Geometric rules must transform consistently under reparameterization relative to the declared subsystem splitting; coordinate mixing that changes that decomposition is a model change. Noisy or stochastic systems require a declared observation scale or a measure designed for stochastic paths. The framework does not claim a unique direct scalar or ledger scalarization at arbitrary resolution. Retrospective restriction also depends on recorded data: a later refinement whose relevant interface or exchange variables were not captured at sufficient resolution cannot be reconstructed and requires a new instrumentation or model lineage.

Lower direct or scalarized accumulated history is not automatically better. A dead or frozen system can have minimal state change while producing no useful output. Optimization must therefore be conditioned on a required function or output. One possible quantity is excess direct state-time: the observed ξ_B^{full} minus the lowest known ξ_B^{full} compatible with the task; analogous ledger-based objectives require an explicitly declared S_B .

9. Discussion

The boundary-first view changes how several familiar engineering statements are interpreted. A stable subsystem may be stable because another subsystem continuously absorbs its disturbance. A low-wear component may be protected by a cooling system whose own history accumulates elsewhere in the ledger. A process may appear energy efficient because maintenance, operator intervention, or rejection of waste heat lies outside the accounting boundary. The compositional claim is stronger than recommending a larger boundary: when the boundary expands at inherited resolution, the previous subsystem description must remain a coherent restriction of the larger one, and direct versus staged expansion must not change the preserved history.

The framework also clarifies the role of environmental design. Active control is sometimes necessary, but repeated correction is not free. If boundary conditions can be changed so that harmful transitions become inaccessible or less likely, the system may require less corrective trajectory while providing the same useful function. This is a design principle rather than a universal law: the environment itself has a cost and must be included when the boundary is expanded.

The framework-definition claim and the usefulness claim should remain separate. The proposal does not depend on weighted arc length: Section 3.10 gives a non-geometric event-count realization, while the geometric example rejects an inadmissible expanded metric before outcome fitting. The Luminus case provides a real-system boundary check in which neighboring-die thermal burden changes from external influence to internal package coupling when the boundary expands, without changing the physical response or inherited self term. These examples establish intelligibility, non-emptiness, and structural discrimination; they do not establish predictive superiority. That empirical question belongs to prospective tests in which a pre-specified M_B or S_B is compared with the strongest established domain method.

Biological and cognitive systems are potential future applications because their internal trajectories depend strongly on environment and nested control. However, such systems introduce major measurement, identifiability, and interpretation problems. The present framework should be validated first in systems with cleaner observability and repeatability.

10. Conclusion

Boundary-relative state-time begins with a simple discipline: define the boundary, state, exchanges, transitions, and accumulation rule before interpreting history. If the analytical boundary later expands at inherited resolution, the contained history and its measurement semantics must remain recoverable; otherwise the larger description is a replacement, not an extension. Coarse-graining and physical modification are separate operations. The geometric and event-count examples show that this requirement is nonempty and not metric-specific, while the Luminus case provides a real coupled-system boundary-reclassification analogue without claiming predictive validation.

The central object is not a universal scalar but a boundary-indexed family of measurement rules, compatible realized histories, component ledgers, and restrictions. Direct and staged restriction must agree, and cross-boundary scalar comparison is allowed only when aggregation semantics also restrict coherently. The mathematical ingredients are established; their assembly here makes one systems claim operational: boundary expansion may reveal additional history and reclassify exchanges without silently rewriting the meaning already assigned to a contained subsystem. The framework earns scientific value only if nontrivial implementations survive prospective comparison with established methods.

Ethics Statement

Not applicable; this theoretical article reports no research involving human participants, animals, biological materials, protected datasets, or non-public sites.

Data Availability Statement

No new primary dataset was generated for this article. Section 4.4 uses publicly released Luminus product and application-note data cited in References [14-16], with Reference [17] providing independent fixture-level context. The values in Table 1 are direct arithmetic decompositions of the published thermal matrix and the manufacturer's published worked-example dissipated powers.

Acknowledgments

During preparation of this manuscript, the author used OpenAI ChatGPT (GPT-5.6; OpenAI, accessed September 2026) for iterative drafting and editing, mathematical and conceptual critique, and assistance identifying literature for subsequent author verification. AI-generated text and suggestions were critically reviewed, revised, and accepted or rejected by the author. All cited sources and calculations were independently checked against primary or authoritative sources. The framework, analysis, interpretations, evidence selection, and conclusions are those of the author; the author retained intellectual control throughout and takes full responsibility for the content and accuracy of the work. AI assistance did not generate empirical data or determine the manuscript conclusions.

References

1. Richardson, I.W.; Rosen, R. Aging and the metrics of time. *Journal of Theoretical Biology* 1979, 79, 415-423. [https://doi.org/10.1016/0022-5193\(79\)90234-0](https://doi.org/10.1016/0022-5193(79)90234-0).
2. Hecksher, T.; Olsen, N.B.; Niss, K.; Dyre, J.C. Physical aging of molecular glasses studied by a device allowing for rapid thermal equilibration. *Journal of Chemical Physics* 2010, 133, 174514. <https://doi.org/10.1063/1.3487646>.

3. Dyre, J.C. Narayanaswamy's 1971 aging theory and material time. *Journal of Chemical Physics* 2015, 143, 114507. <https://doi.org/10.1063/1.4930551>.
4. Douglass, I.M.; Dyre, J.C. Distance-as-time in physical aging. *Physical Review E* 2022, 106, 054615. <https://doi.org/10.1103/PhysRevE.106.054615>.
5. Crooks, G.E. Measuring thermodynamic length. *Physical Review Letters* 2007, 99, 100602. <https://doi.org/10.1103/PhysRevLett.99.100602>.
6. Hollerbach, R.; Dimanche, D.; Kim, E.-J. Information Geometry of Nonlinear Stochastic Systems. *Entropy* 2018, 20, 550. <https://doi.org/10.3390/e20080550>.
7. Jacquet, Q.; Kim, E.-J.; Hollerbach, R. Time-Dependent Probability Density Functions and Attractor Structure in Self-Organised Shear Flows. *Entropy* 2018, 20, 613. <https://doi.org/10.3390/e20080613>.
8. Gu, J. Thermodynamic bounds on the asymmetry of cross-correlations with dynamical activity and entropy production. *Physical Review E* 2024, 109, L042101. <https://doi.org/10.1103/PhysRevE.109.L042101>.
9. Nagayama, R.; Yoshimura, K.; Ito, S. Infinite variety of thermodynamic speed limits with general activities. *Physical Review Research* 2025, 7, 013307. <https://doi.org/10.1103/PhysRevResearch.7.013307>.
10. Hasegawa, Y. Thermodynamic Correlation Inequality. *Physical Review Letters* 2024, 132, 087102. <https://doi.org/10.1103/PhysRevLett.132.087102>.
11. Mac Lane, S.; Moerdijk, I. *Sheaves in Geometry and Logic: A First Introduction to Topos Theory*. Springer-Verlag: New York, NY, USA, 1992. <https://doi.org/10.1007/978-1-4612-0927-0>.
12. Fong, B.; Spivak, D.I. *An Invitation to Applied Category Theory: Seven Sketches in Compositionality*. Cambridge University Press: Cambridge, UK, 2019. <https://doi.org/10.1017/9781108668804>.
13. Bhapkar, V.P.; Srinivasan, C. On Fisher information inequalities in the presence of nuisance parameters. *Annals of the Institute of Statistical Mathematics* 1994, 46, 593-604. <https://doi.org/10.1007/BF00773520>.
14. Luminus Devices, Inc. CXM-14 Product Datasheet, Generation 4; PDS-003005 Rev. 04; Luminus Devices, Inc., 2025. Available online: https://download.luminus.com/datasheets/Luminus_CXM-14_GEN4_Datasheet.pdf (accessed 5 September 2026).
15. Luminus Devices, Inc. Thermal Management of Big Chip LEDs; APN-001443 Rev. 04; Luminus Devices, Inc., 2013. Available online: https://download.luminus.com/datasheets/Luminus_APN_APN-001443.pdf (accessed 5 September 2026).
16. Luminus Devices, Inc. Thermal Cross-Talk in Multi-Chip LED Packages; APN-002127 Rev. 03; Luminus Devices, Inc., 2020. Available online: https://download.luminus.com/datasheets/Luminus_APN_APN-002127.pdf (accessed 5 September 2026).
17. Kusuma, P.; Pattison, P.M.; Bugbee, B. From physics to fixtures to food: current and potential LED efficacy. *Horticulture Research* 2020, 7, 56. <https://doi.org/10.1038/s41438-020-0283-7>.
18. Valanis, K.C. A theory of viscoplasticity without a yield surface. Part I: General theory. *Archives of Mechanics* 1971, 23, 517-533.
19. Valanis, K.C.; Lee, C.-F. Some recent developments of the endochronic theory with applications. *Nuclear Engineering and Design* 1982, 69, 327-344. [https://doi.org/10.1016/0029-5493\(82\)90181-9](https://doi.org/10.1016/0029-5493(82)90181-9).
20. Sapareto, S.A.; Dewey, W.C. Thermal dose determination in cancer therapy. *International Journal of Radiation Oncology, Biology, Physics* 1984, 10, 787-800. [https://doi.org/10.1016/0360-3016\(84\)90379-1](https://doi.org/10.1016/0360-3016(84)90379-1).
21. van der Schaft, A.J.; Maschke, B.M. Hamiltonian formulation of distributed-parameter systems with boundary energy flow. *Journal of Geometry and Physics* 2002, 42, 166-194. [https://doi.org/10.1016/S0393-0440\(01\)00083-3](https://doi.org/10.1016/S0393-0440(01)00083-3).

22. Heijungs, R.; Allacker, K.; Benetto, E.; Brandão, M.; Guinée, J.; Schaubroeck, S.; Schaubroeck, T.; Zamagni, A. System Expansion and Substitution in LCA: A Lost Opportunity of ISO 14044 Amendment 2. *Frontiers in Sustainability* 2021, 2, 692055. <https://doi.org/10.3389/frsus.2021.692055>.
23. Willems, J.C. The Behavioral Approach to Open and Interconnected Systems. *IEEE Control Systems Magazine* 2007, 27, 46-99. <https://doi.org/10.1109/MCS.2007.906923>.
24. Fong, B. Decorated Cospans. *Theory and Applications of Categories* 2015, 30, 1096-1120. <https://doi.org/10.70930/tac/8cw8vm2>.
25. Baez, J.C.; Courser, K. Structured Cospans. *Theory and Applications of Categories* 2020, 35, 1771-1822. <https://doi.org/10.70930/tac/dca3obx4>.
26. Baez, J.C.; Courser, K. Coarse-Graining Open Markov Processes. *Theory and Applications of Categories* 2018, 33, 1223-1268. <https://doi.org/10.70930/tac/14gq63yh>.
27. Robinson, M. Sheaves are the canonical data structure for sensor integration. *Information Fusion* 2017, 36, 208-224. <https://doi.org/10.1016/j.inffus.2016.12.002>.
28. Wittner, R.; Holub, P.; Mascia, C.; et al. Toward a common standard for data and specimen provenance in life sciences. *Learning Health Systems* 2024, 8, e10365. <https://doi.org/10.1002/lrh2.10365>.
29. Putnick, D.L.; Bornstein, M.H. Measurement invariance conventions and reporting: The state of the art and future directions for psychological research. *Developmental Review* 2016, 41, 71-90. <https://doi.org/10.1016/j.dr.2016.06.004>.
30. Joint Committee for Guides in Metrology (JCGM). International Vocabulary of Metrology - Basic and General Concepts and Associated Terms (VIM), 3rd ed.; JCGM 200:2012; BIPM: Sèvres, France, 2012. Available online: https://www.bipm.org/documents/20126/2071204/JCGM_200_2012.pdf (accessed 5 September 2026).
31. Osejo-Bucheli, C. Temporal Viability Theory: A Cybernetic Framework for Time-Bound Viability in Socio-Technical and Algorithmic Systems. *Systems Research and Behavioral Science* 2026, 43, 1364-1379. <https://doi.org/10.1002/sres.70036>.
32. Dynkin, E.B. Additive functionals of Markov processes and stochastic systems. *Annales de l'Institut Fourier* 1975, 25, 177-200. <https://doi.org/10.5802/aif.578>.
33. Chen, Z.-Q.; Fukushima, M. *Symmetric Markov Processes, Time Change, and Boundary Theory*; Princeton University Press: Princeton, NJ, USA, 2011. <https://doi.org/10.23943/princeton/9780691136059.001.0001>.
34. Vilela Mendes, R. Projective Limits and Consistent Measures in Physical Problems. *International Journal of Theoretical Physics* 2026, 65, 199. <https://doi.org/10.1007/s10773-026-06400-7>.
35. Alur, R.; Grosu, R.; Lee, I.; Sokolsky, O. Compositional modeling and refinement for hierarchical hybrid systems. *The Journal of Logic and Algebraic Programming* 2006, 68, 105-128. <https://doi.org/10.1016/j.jlap.2005.10.004>.
36. D'Argenio, P.R.; Verhoef, C. A general conservative extension theorem in process algebras with inequalities. *Theoretical Computer Science* 1997, 177, 351-380. [https://doi.org/10.1016/S0304-3975\(96\)00292-7](https://doi.org/10.1016/S0304-3975(96)00292-7).